

screens and sliders. Users are able to communicate with their appliances better via the Internet. Capacitive touch screens driven by IoT give a slew of benefits to appliance manufacturers including:

1. Remote maintenance
2. Better understanding of usage with data mining and analytics, which can lead to product improvements or customisation
3. Power management—syncing high current appliances to avoid spikes on the power line
4. Firmware updates for remote bug fixes and product enhancements

But IoT enabled home and kitchen appliances can also benefit the end-user. Manufacturers, for instance, can provide:

1. 'How to use' instructions for some appliances, or recipes and even some best practices that could be by way of videos or easy downloads. Laundry appliances such as washing machines or dryers could come with their own instructions on usage of washing machine powder, type of washing powder or liquid and quick troubleshooting instructions. Firmware updates to add new features and improve performance.

2. Weather, news and other updates based on user preferences.

However, with all of these benefits for suppliers and users, there are a few design issues that must be addressed.

Addressing white goods application issues

Appliances that are fitted with touch screen HMIs are challenged when it comes to noise, moisture or recognising touch screen commands from a user wearing gloves. If manufacturers of these appliances can address some of these issues with their IC or touch screen controller, then their products will definitely have the edge.

Noise. For noise immunity, patented technology allows the controller to be more immune to the noise coming over the power lines. In several areas, power lines generate a lot of noise because of the lack of an earth ground or poorly connected grounds. That noise conducts into the power supply through the power cable and gets routed to the touch controller IC.



Fig. 2: Behind the touch screen, an array of drive electrodes receives electrodes and the circuitry in the touch IC faithfully detects a user's touch

Being an extremely sensitive component that measures nanocoulombs of charge, pulling even a tiny amount of charge away by simply touching the touch screen with a finger needs to be interpreted consistently and accurately. Noise can inject significant charge into the sensor and hence a touch controller without sufficient noise immunity can lead to confusion in the interpretation of signals, which is critical to the proper functioning of the appliance. In case of a ghost touch or false touch event, there's an erratic behaviour of buttons where they start pressing themselves randomly. Imagine if you are using an oven and this were to happen, it can have dangerous consequences. The same randomness could affect other appliances and lead to unstable and dangerous scenarios. This is an issue for every capacitive touch controller, but patented technology circumvents conducted noise and can help control the issue.

To solve the noise problem, the controller filters common-mode noise and avoids noise issues through a frequency hopping scheme. The patented approach utilises self-capacitive touch as well as mutual capacitive touch scanning and involves differential touch sensing. Instead of treating each sense line as its own individual element, the IC measures the difference between pairs of sense lines and is able to eliminate noise common

to both sense lines. If, for instance, the same noise occurs from a similar region of the display, it is cancelled, retaining only the valid signal. This differential touch sensing provides a very effective level of noise cancellation/rejection.

Washing machine motors, refrigerator compressors or inductive cooktop burners radiate noise that is within the acceptable noise cancelling frequency band, and hence touch screen displays for these appliances are more robust. False touch events are thus avoided in these instances. However, in the case of a legitimate touch, it must be reported and detected to the host controller to avoid a missed touch event from noise, so that user expectation of a response is met.

Moisture. It is natural for moisture and wetness to occur in laundry rooms and kitchens, which are a necessity while cooking or washing. Imagine boiling water for tea and it spills over on to the touch screen, a false event could also occur. Mist, droplets of water, liquids of any kind could affect the responsiveness and functionality of the touch screen. This means, touch screen designers of HMI appliances need to create designs that can detect a touch despite water or moisture. Multi-touch operations, from the standard two touches to perhaps ten or more by multiple users at the same time, must be made avail-